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ASSIGNMENT - 9 CHE 312

Q1

Net rate $= R_{AB} + R_{AF} = -(k_B + k_F) = -k_T$

$$\boxed{k_T = k_B + k_F}$$

$$\frac{dN_A}{dy} = r_A$$

$$N_A = -D_A \frac{dC_A}{dy}$$

$$\frac{d}{dy}\left(-D_A \frac{dC_A}{dy}\right) = -k_T \Rightarrow \frac{d}{dy}\left(D_A \frac{dC_A}{dy}\right) = k_T$$

$$D_A = \frac{D_{A0}}{1+\alpha y}$$

$$\frac{d}{dy}\left(\frac{D_{A0}}{1+\alpha y} \frac{dC_A}{dy}\right) = k_T$$

$$\frac{D_{A0}}{1+\alpha y} \frac{dC_A}{dy} = k_T y + C_1$$

$$\frac{dC_A}{dy} = \frac{1+\alpha y}{D_{A0}} (k_T y + C_1)$$

$y = L \Rightarrow$ no flux

$$N_A(L) = -D_A(L) \left.\frac{dC_A}{dy}\right|_{y=L} = 0 \Rightarrow \left.\frac{dC_A}{dy}\right|_{y=L} = 0$$

$$\frac{1+\alpha L}{D_{A0}}(k_T L + C_1) = 0$$

$$k_T L + C_1 = 0 \qquad \Rightarrow C_1 = -k_T L$$

$$\frac{dC_A}{dy} = \frac{1+\alpha y}{D_{A0}}[k_T y - k_T L] = \frac{k_T}{D_{A0}}(1+\alpha y) * (y-L)$$

$$C_A(y) = \frac{k_T}{D_{A0}}\int (1+\alpha y)(y-L)\,dy + C_2$$

$$(1+\alpha y)(y-L) = y - L + \alpha y(y-L)$$

$$= y - L + \alpha y^2 - \alpha L y$$

$$= \alpha y^2 + (1-\alpha L)y - L$$

$$C_A(y) = \frac{k_T}{D_{A0}}\left[\alpha\frac{y^3}{3} + (1-\alpha L)\frac{y^2}{2} - Ly\right]$$

$$C_A(0) = C_0 \quad \& \quad \left.\frac{dC_A}{dy}\right|_{y=L} = 0$$

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$$-r_A = k_T = k_B + k_F$$

$$R_{overall} = \int_0^L (k_B + k_F)\,dy = (k_B + k_F)L$$

$y = 0$:

$$N_A(0) = -D_A(0)\left.\frac{dC_A}{dy}\right|_{y=0} = (k_B + k_F)L$$

$$R_{overall} = (k_B + k_F)L$$

Q2

$$Q = m\,C_{p,h}(T_{h,i} - T_{h,o}) = 600(110 - 80) = 1.8 \times 10^4\ W$$

$$Q = m\,C_{p,h}(T_{c,o} - T_{c,i}) \Rightarrow 1.8 \times 10^4 = 800(T_c - 30)$$

$$T_{c,o} - 30 = \frac{1.8 \times 10^4}{800} = 22.5 \Rightarrow T_{c,o} = 52.5\,^\circ C$$

Counter Current

$$\Delta T_1 = T_{h,i} - T_{c,o} = 110 - 52.5 = 57.5\,^\circ C$$

$$\Delta T_2 = T_{h,o} - T_{c,i} = 80 - 30 = 50\,^\circ C$$

$$\Delta T_{lm,cc} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)} = \frac{57.5 - 50}{\ln\left(\frac{57.5}{50}\right)}$$

$$= 53.7\ k$$

$$A_{cc} = \frac{Q}{U\Delta T_{lm,cc}} = \frac{1.8 \times 10^4}{400 \times 53.7} = 0.84\ m^2$$

Co Current

$\Delta T_1 = 110 - 30 = 80°C$

$\Delta T_2 = 80 - 52.5 = 27.5°C$

$$\Delta T_{lm,Co} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)} = \frac{80 - 27.5}{\ln(80/27.5)} = 49.2 K$$

$$A_{Co} = \frac{Q}{U \Delta T_{lm,Co}} = \frac{1.8 \times 10^4}{400 \times 49.2} = 0.92 m^2$$